Sparse Autoencoders for Interpretable Latent Factor Models in Asset Pricing

Project Proposal

Justin Tarquino and Shreya Enaganti

1. Goals and Motivation

Asset pricing models such as CAPM (Capital Asset Pricing Model) and the Fama–French framework explain the cross-section of stock returns using a small number of factors. Over the past decade, more than 150 candidate factors have been documented in the empirical finance literature, prompting the question of how to compress this "factor zoo" into a concise model. Recent work has applied PCA, autoencoders (AE), and variational autoencoders (VAE) to compress these factors into a small set of latent factors with strong pricing performance. However, AE-based factor models remain difficult to interpret because each latent factor is typically a dense nonlinear combination of all inputs. This project studies the tradeoff between interpretability and pricing performance in sparse AE-based latent factor models. Specifically, we ask:

- How does enforcing sparsity in latent factor loadings affect cross-sectional pricing performance?
- And does this sparsity–performance tradeoff differ between standard autoencoders, denoising autoencoders (DAEs), and variational autoencoders (VAEs)?

Interpretability is increasingly important in applied finance because regulators and risk managers need economically understandable models, not only predictive accuracy. While sparse PCA factors have been shown to closely approximate dense PCA factors, it is unclear whether similar results hold in nonlinear AE settings.

2. Related Work

Lettau & Pelger (2019) propose Risk-Premium PCA (RP-PCA), which incorporates pricing information into factor extraction. Gu, Kelly & Xiu (2021) and Ben Chaouch et al. (2023) demonstrate that autoencoder-based factor models outperform linear baselines in pricing tasks. Feiler, M. J. (2024) maintains the AE approach as a strong denoiser of financial time series. Pelger & Xiong (2018) obtain sparse interpretable PCA factors through post-hoc shrinkage, but do not study nonlinear AE factors or pricing performance under sparsity constraints. Denoising autoencoders have also been shown to produce more robust representations, though their interaction with sparsity in asset pricing remains unexplored.

3. Data and Methods

We will use the 150-factor dataset from Feng, Giglio & Xiu (2020), consisting of monthly data from 1976–2017, where each factor represents the return of a portfolio based on firm characteristics such as value, momentum, size, and profitability. We will also use standard Fama–French benchmark test portfolios for evaluation. Models will be implemented in PyTorch, with PCA and RP-PCA baselines in scikit-learn. We will implement autoencoder models with L_1 regularization on encoder weights to enforce sparsity, alongside L_2 regularized and denoising autoencoder variants for comparison. The learned latent representations z_t from each model (and latent distribution for the VAE) will be used (sampled from for VAE) as factors in a standard cross-sectional asset pricing framework. Models will be evaluated using a rolling-window framework from 1976-2017. For each window, models are trained on the data from a certain N-year long period, hyperparameters are tuned on the subsequent M-year long validation period, and performance is evaluated on the following K-year long test period. N, M, and K will aslo be tuned based on results. Performance will be measured using pricing-error metrics (like MSE, R^2) while sparsity will be measured via the fraction of near-zero encoder weights.

4. Planned Experiments

1. Measure the sparsity-performance frontier for AE, DAE, and VAE models across varying penalties and architectures
2. Compare nonlinear sparse models against sparse RP-PCA and PCA baselines
3. Analyze whether sparse latent factors align with known economic themes such as value, momentum, and profitability

5. Timeline and Resources

The project timeline consists of:

- Week 7: data preprocessing and AE baseline validation,
- Week 8: sparse AE, DAE, VAE experiments,
- Week 9-10: RP-PCA baselines, interpretation analysis, and final report preparation.

The computational requirements are modest and we expect the full experimental grid is expected to require only a few GPU-hours on the TTIC compute cluster.

6. Anticipated Division of Labor

One group member will focus on data preprocessing and baseline model implementation (PCA and RP-PCA), while the other will focus on autoencoder development and sparsity regularization; both will collaborate on evaluation, backtesting, and analysis of sparsity–performance tradeoffs.

7. References

1. Ben Chaouch, Z., Lo, A. W., Singh, M., & Xiong, K. (2023). *Chimeras in the Factor Zoo*. MIT CSAIL/LFE Working Paper.

2. Feng, G., Giglio, S., & Xiu, D. (2020). Taming the factor zoo: A test of new factors. *Journal of Finance*, 75(3), 1327–1370.

3. Gu, S., Kelly, B., & Xiu, D. (2021). Autoencoder asset pricing models. *Journal of Econometrics*, 222(1), 429–450.

4. Lettau, M., & Pelger, M. (2019). Estimating latent asset-pricing factors. *Journal of Econometrics*, 218(1), 1–31.

5. Lettau, M., & Pelger, M. (2018). Factors that fit the time series and cross-section of stock returns. *Review of Financial Studies*, 33(5), 2274–2325.

6. Pelger, M., & Xiong, R. (2018). Interpretable sparse proximate factors for large dimensions. *Journal of Business & Economic Statistics*, 40(4), 1642–1664.

7. Vincent, P., Larochelle, H., Bengio, Y., & Manzagol, P.-A. (2008). Extracting and composing robust features with denoising autoencoders. *ICML*.

8. Duan, Y., Wang, L., Zhang, Q., & Li, J. (2022). FactorVAE: A probabilistic dynamic factor model based on variational autoencoder for predicting cross-sectional stock returns. *Proceedings of the AAAI Conference on Artificial Intelligence*, 36(4), 4468–4476.

9. Feiler, M. J. (2024). Strong denoising of financial time-series. arXiv preprint arXiv:2408.05690.